VIZIJ AFTERNOON WORKSHOP

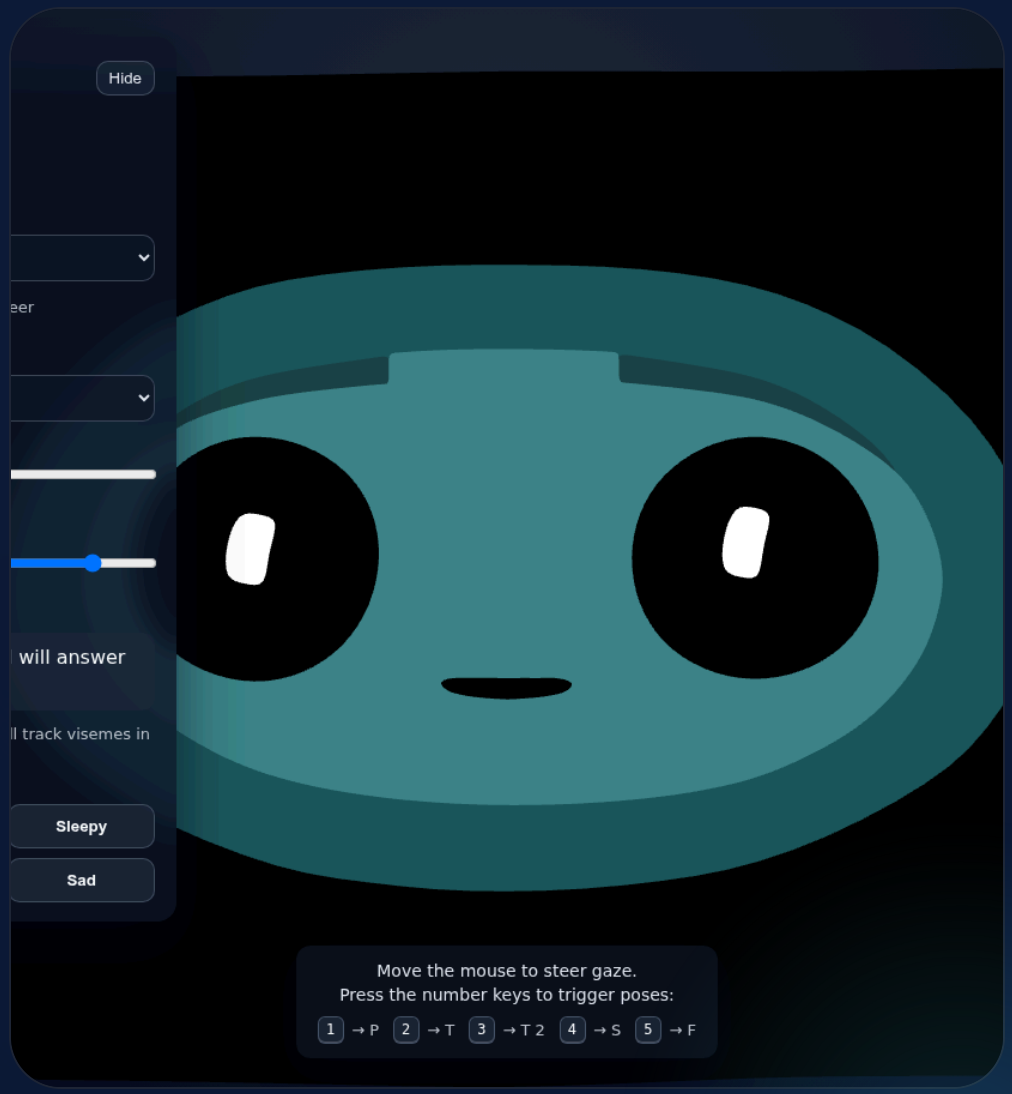
Design, Animate, and Deploy Robot Faces with Vizij

A hands-on session built around the current Vizij authoring, runtime, animation, and deployment surfaces.

3.5 hrs
tutorial + breakout flow

5 apps
current demo anchors

2 labs
plus roadmap session



Capstone teaser: live face runtime with gaze, emotions, and conversation controls.

OUTCOMES

What Participants Leave Able To Do

1. Load and control a Vizij face in a React runtime.
2. Inspect the renderer data model and trace control paths.
3. Use `vizij-authoring` to reason about rigging and poses.
4. Understand stored animation versus live runtime control.
5. Explain the path from local demo to robot-facing deployment.
6. Contribute grounded roadmap and strategy feedback.

Teaching rhythm

Orient · Demo · Try · Debrief

Every block is structured around a visible success check, not a lecture-only explanation.

AGENDA

Run of Show

2:30 pm	Tutorial introduction and capstone teaser
2:45 pm	Activity 1: face design, Hello Face, renderer data model, rigging
3:30 pm	Technical support or exploration time
3:45 pm	Activity 2: animation, integration, deployment
4:30 pm	Technical support or exploration time
4:45 pm	Coffee break
5:00 pm	Roadmap, research, and strategy breakouts
6:00 pm	Tutorial concludes

ARCHITECTURE

How Today’s Apps Fit Together

Authoring

`vizij-authoring`

Import, inspect, organize, and export face assets and rig data.

Runtime

`tutorial-fullscreen-face`

Smallest useful runtime-react face with gaze and pose hotkeys.

Animation

`demo-vizij-player`

Blend rigs, trigger clips, and inspect orchestrator diagnostics.

Conversation

`tutorial-agent-face`

Live visemes, emotion tools, and function-called gaze control.

Deployment

`vizij-standalone`

Desktop runtime with WebSocket control and a built-in control panel.

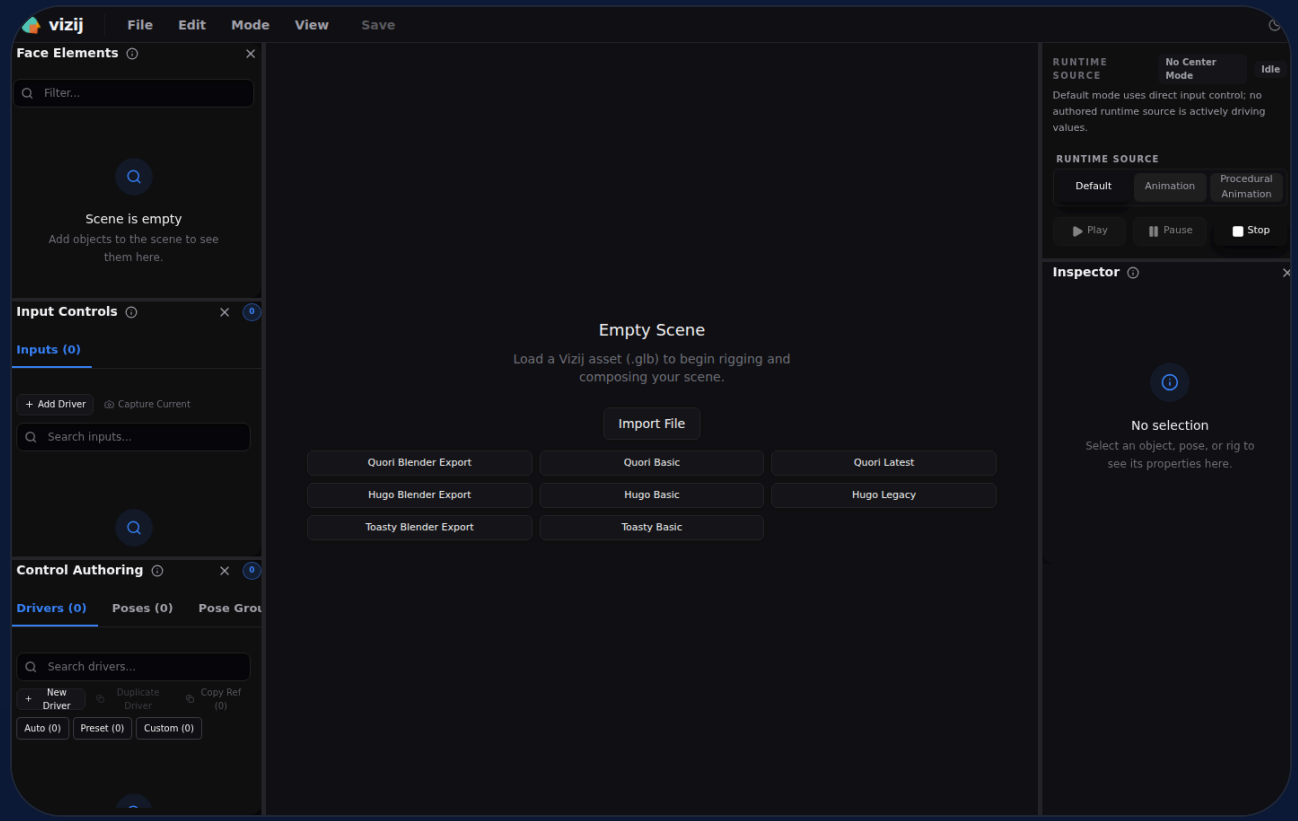
Strategy

`vizij-docs/ROADMAP`

Translate workshop experience into informed product direction.

ACTIVITY 1

Face Design and Control Starts in Authoring

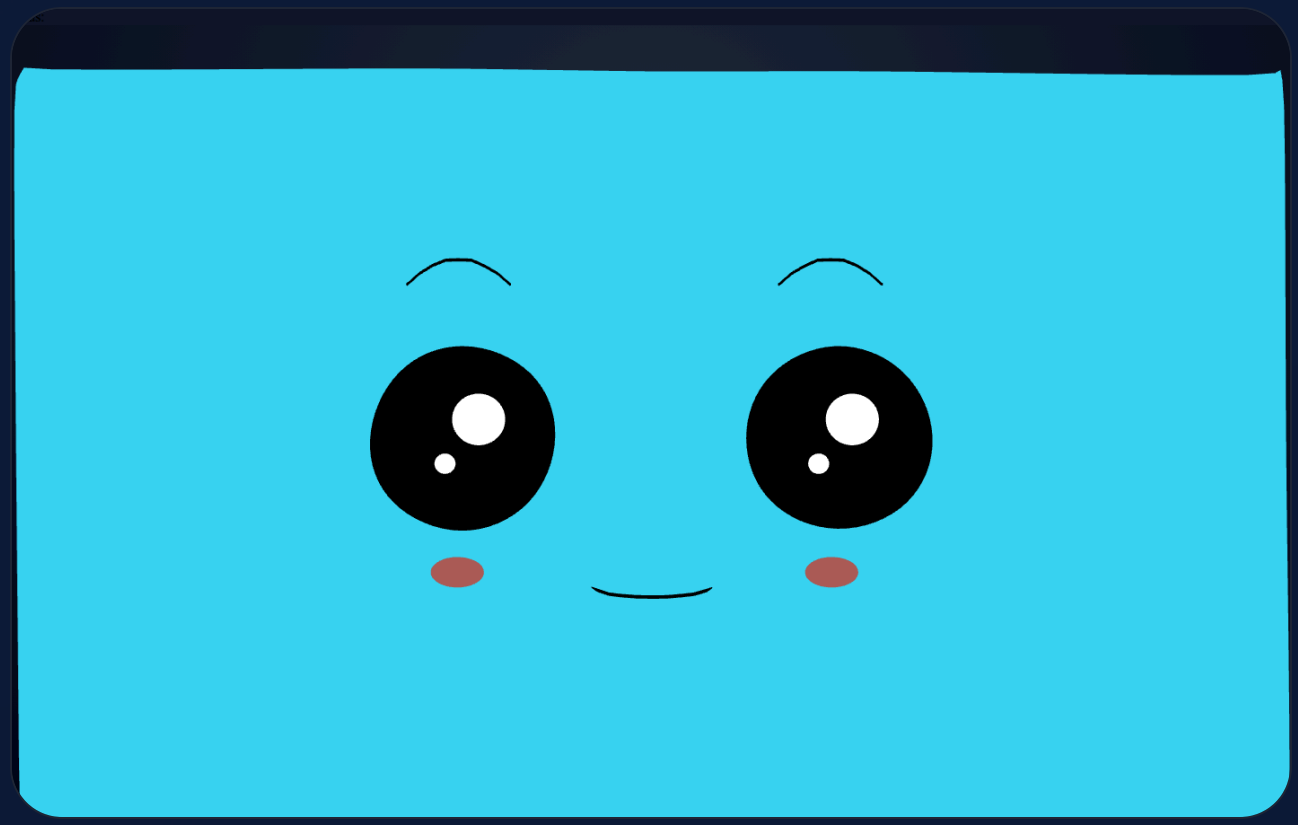


- Load a sample GLB or bundled face.
- Identify standard input routes and visible animatable features.
- Explain the difference between a standard control and a pose weight.
- Use the empty-scene entry points to keep setup fast in the room.

Presenter note: the empty-state sample buttons are useful for workshops because they remove file-picking friction.

HELLO FACE

Runtime React in the Small



- `VizijAssetBundle` defines the face and metadata.
- `VizijRuntimeProvider` owns loading and orchestration.
- `VizijRuntimeFace` renders the visual surface.
- Mouse gaze and pose hotkeys prove the path wiring quickly.

Workshop checkpoint: participants should be able to point at one path and explain what visible change it causes.

RENDERER DATA MODEL

What Changes When You Set a Value

- Inputs are staged into the orchestrator.
- Resolved paths include the face id and namespace.
- Outputs flow into the renderer store and change the face.

```
rig/{face}/standard/left_eye/pos/x
rig/{face}/standard/right_eye/pos/y
rig/{face}/poses/{pose_name}.weight
```

These three path shapes are enough to explain most of the workshop’s visible control flows.

RIGGING

Standard Inputs Give You Reuse, Pose Weights Give You Expression

- Standard inputs make different faces interoperable.
- Pose groups are the fastest workshop-friendly abstraction.
- Authoring is the place to inspect naming before runtime code.

standard/left_eye/pos/x

standard/mouth/...

poses/smile.weight

visemes/...

Suggested teaching move: ask participants which controls should be universal versus face-specific.

SUPPORT WINDOW

Keep the Room Moving

Lane 1

Install + env

Recover setup blockers without stopping the main flow.

Lane 2

Path confusion

Use the handouts to reconnect visuals to path patterns.

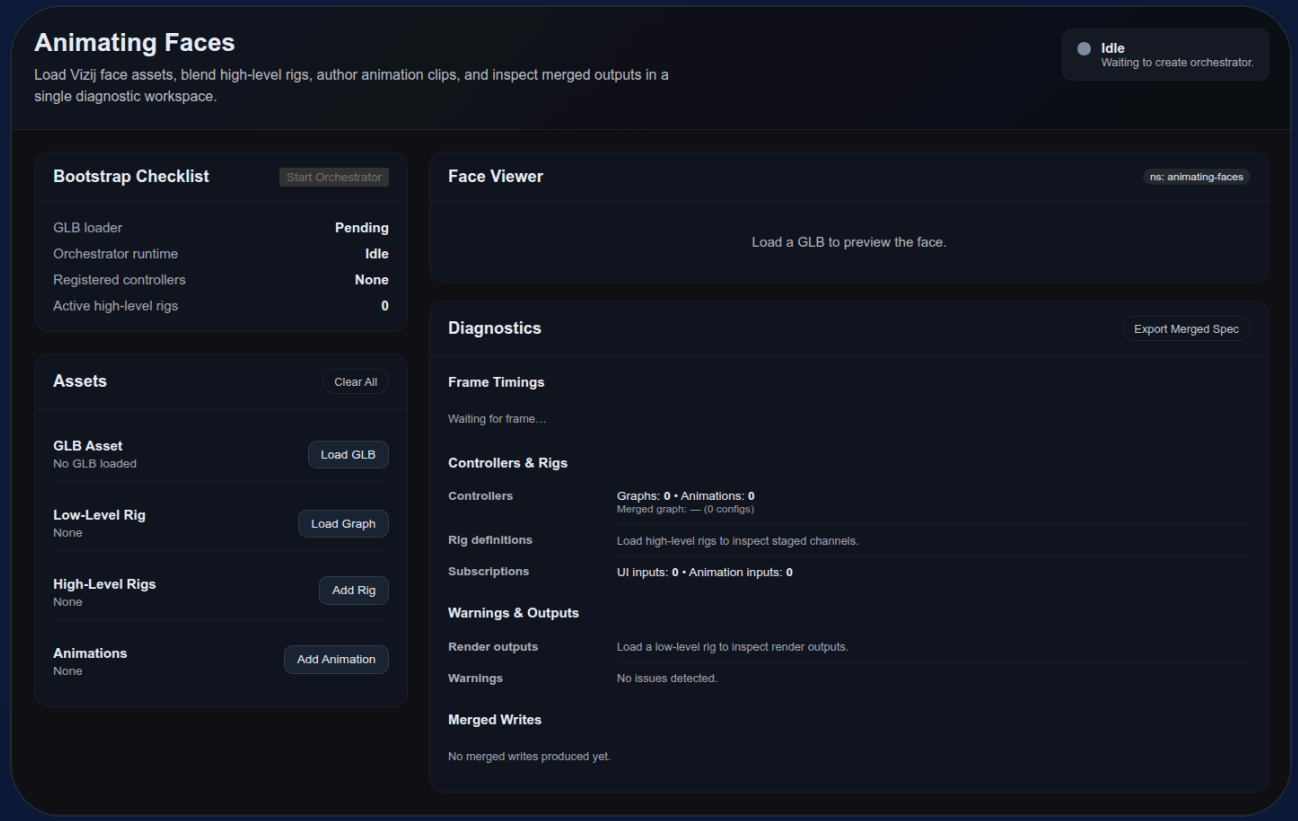
Lane 3

Stretch tasks

Give fast participants exploration prompts instead of idle time.

ACTIVITY 2

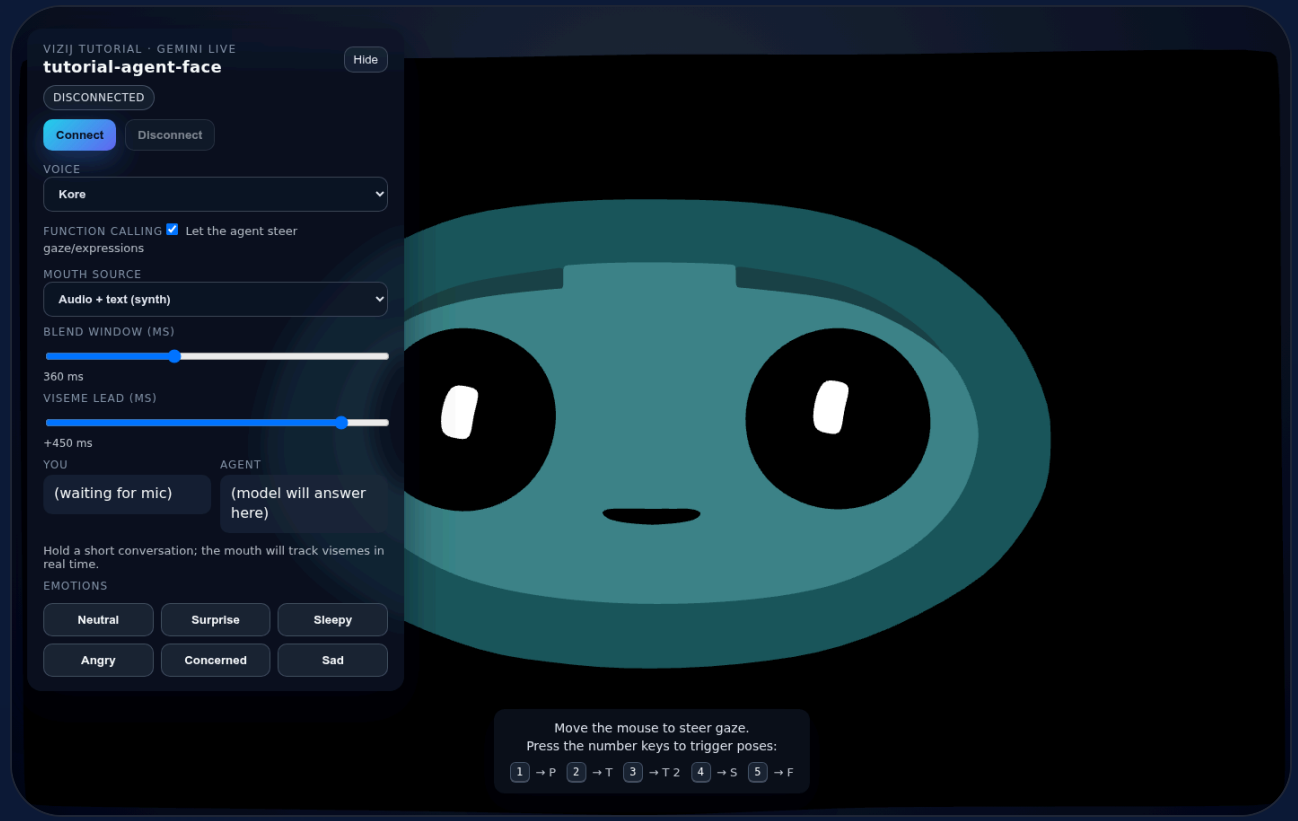
Animation and Orchestration



- Use the panel layout to explain assets, controls, and diagnostics.
- Describe how manual rig inputs and stored animation can both contribute.
- Keep this as the backup concept anchor if live conversation is unavailable.

CONVERSATION

Live Control Adds Timing, Audio, and Emotion



- Mouth source modes let you compare baseline, synth, and align strategies.
- Emotion buttons and function-calling show non-verbal behavior hooks.
- This segment depends on `VITE_GEMINI_API_KEY`, so treat it as facilitator-led by default.

Good workshop move: teach the control model even if the room is not fully credentialed.

DEPLOYMENT

From Laptop Demo to Robot-Facing Runtime

- `vizij-standalone` packages a runtime plus WebSocket control surface.
- The face stays the same; the control layer changes.
- Typed slot values make external control explicit and predictable.

```
{
  "type": "set_slot_values",
  "values": {
    "rig/robot/standard/left_eye/pos/x": { "f64": 0.5 }
  }
}
```

Use the README walkthrough and message format instead of trying to force a full Tauri build for every participant.

BREAKOUTS AND CLOSE

Turn the Hands-On Session into a Next-Step Decision

1. Product and roadmap priorities
2. Research and technical opportunities
3. Deployment and adoption risks

- Each group reports one opportunity.
- Each group reports one blocker.
- Each group proposes one next experiment or milestone.

Close with the question: what can you now control, explain, or ship more confidently than you could at 2:30 pm?